Back-of-the-Envelope Estimation (Twitter QPS + Storage Example)

Back-of-the-envelope estimation is a **core system design interview skill** used to approximate:

- Traffic (QPS)
- Storage requirements
- Server capacity
- Cache size

The goal is **not exact numbers**, but a clear **logical estimation process**.

1. Problem Statement (Example: Twitter)

We estimate system scale based on assumptions (not real Twitter data).

Assumptions

```
None

• Monthly Active Users (MAU) = 300 million

• 50% users are daily active → DAU = 150 million

• Each user posts = 2 tweets/day

• 10% tweets contain media

• Data retention = 5 years
```

2. QPS (Queries Per Second) Estimation

Step 1: Daily Tweet Volume

```None
DAU = 300M × 50% = 150M users
```

```None
Tweets per day = 150M × 2 = 300M tweets/day
```

Step 2: QPS Calculation

There are 86,400 seconds in a day:

```None
QPS = 300M / 86400 ≈ 3500 QPS
```

Step 3: Peak QPS

Traffic is not uniform → assume peak = 2× average:

```None
Peak QPS ≈ 7000 QPS
```

3. Storage Estimation (Media Only)

We only calculate **media storage (images/videos)**.

Step 1: Media Tweets per Day

```None
Media tweets/day = 300M × 10% = 30M tweets/day
```

Step 2: Average Media Size

```None
Tweet ID  → 64 bytes
Text      → 140 bytes
Media     → ~1 MB (dominant factor)
```

We ignore small fields since media dominates storage.

Step 3: Daily Storage

```None
Daily media storage = 30M × 1 MB = 30 TB/day
```

Step 4: 5-Year Storage

```None
5 years = 365 × 5 = 1825 days
```

```None
Total storage = 30 TB × 1825 ≈ 55 PB
```

4. Key Interview Insights

✔ 1. Approximation is expected

```None
100,000 / 9.1 ≈ 100,000 / 10 ≈ 10,000
```

Precision is NOT important in system design interviews.

✔ 2. Always write assumptions

Makes your reasoning traceable and structured.

✔ 3. Always mention units

```None
Wrong → 30

Correct → 30 TB/day
```

Avoid ambiguity in system design answers.

✔ 4. Break problem into steps

Typical flow:

```None
Users → Requests → QPS → Storage → Servers
```

5. Common Estimation Questions in Interviews

You should practice:

- QPS (average + peak)
- Storage (daily + yearly)
- Cache size
- Number of servers required
- Bandwidth estimation

6. System Design Mindset

```None
System design = estimation + trade-offs + bottleneck handling
```

Estimation helps identify system constraints early.

7. One-Line Summary

Back-of-the-envelope estimation is a structured approximation technique used to calculate system scale (QPS, storage, bandwidth) using simplified assumptions, rounding, and step-by-step breakdown.